

# Database Design and Development

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## Computing and Mathematics

May, 2026

# Database Design and Development (A33961)

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|---------------------|---------------------------|------------------------|
| <b>Short Title:</b> | DB Design and Development |                        |
| <b>Faculty</b>      | Science and Computing     |                        |
| <b>Department</b>   | Computing and Mathematics |                        |
| <b>Cluster</b>      | Database and Analytics    |                        |
| <b>Author</b>       | RSHANAHAN                 |                        |
| <b>Credits</b>      | 5                         | <b>Level:</b> Advanced |

## Description of Module / Aims

This module will introduce the student to the principles and practice of designing and implementing efficient and secure database systems. The student will gain competence in designing relational databases using Entity Relationship Modelling. They will implement relational databases using SQL data definition language. They will query the relational database using SQL data manipulation language. The students will be introduced to the concepts and use of NoSQL databases.

## Programmes

|           |                                                                     |           |
|-----------|---------------------------------------------------------------------|-----------|
| DATA-0018 | Higher Diploma in Science in Business Systems Analysis (WD_KBUSY_G) | 1 / 1 / M |
|-----------|---------------------------------------------------------------------|-----------|

## Indicative Content

- Database Environment; Relational Database Analysis and Design
- ER modelling of DB concepts
- Development of Conceptual model to Logical model
- Physical Database Design;
- Database Implementation
- Database Security & Data Protection
- Introduction to NoSQL Databases

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Appraise the role of a database and its management system.
2. Create Entity Relationship (ER) diagrams from business scenarios and reproduce those diagrams into normalised relations ready for database implementation.
3. Construct a physical database design.
4. Implement a relational database using SQL Data Definition Language (DDL).
5. Construct queries on a relational database using SQL Data Manipulation Language (DML).
6. Investigate Database Security Procedures; Introduce concepts of Data Protection (DGPR, CCPA).
7. Introduce NoSQL Databases

## Learning and Teaching Methods

- The lectures will introduce the theory content to the student. The student will be encouraged to participate in class discussions and ask questions to support their learning process.
- The practical classes facilitate the student in implementing the theory learned in the lectures using incremental steps to accomplish a skill.
- The continuous assessment will require the student to apply the theory and practical knowledge to a new business scenario.

## Assessment Methods

|                            | Weighing | Outcomes Assessed |
|----------------------------|----------|-------------------|
| Continuous Assessment      | 100%     | 1,2,3,4,5,6,7     |
| <i>Assignment</i>          | 55%      | 2,3               |
| <i>Assignment</i>          | 30%      | 4,5               |
| <i>In-Class Assessment</i> | 15%      | 1,6,7             |

## Assessment Criteria

- <40%:** Unable to interpret and describe key concepts of the Database design and implementation domain.
- 40%-49%:** Be able to interpret and describe key concepts of the Database design and implementation domain, particularly the use of ER diagrams for the design of relational databases and SQL DDL for the implementation of the relational database.
- 50%-59%:** Ability to discuss key concepts of the Database design and implementation domain and ability to discover and integrate related knowledge in other knowledge domains. To design and develop a suitable relational database system for a business scenario.
- 60%-69%:** Be able to solve basic business analytics problems by experimenting with the appropriate skills and tools. To design and develop a suitable relational database system for a business scenario to a level that includes an enhanced ER diagram and suitable queries appropriate to the business scenario.
- 0%-100%:** All the above to an excellent level. Be able to analyse, design and implement solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Lecture              | 24        | 24        |
| Practical            | 24        | 24        |
| Independent Learning | 87        | 87        |

## Essential Material

- Connolly, T. and C. Begg. \_Database Systems: A Practical Approach to Design Implementation and Management\_. 6th. NY: Adison Wesley, 2019.

## Supplementary Material

- Harrington, J. \_Relational Database Design and Implementation: Clearly Explained\_. 4th ed. London: Morgan Kaufmann, 2016.

## Requested Resources

- COMPUTER LAB: BYOD Lab